GENERAL SCIENCE

WEEK 3

SIMPLE AND COMPOUND MACHINES

NAME OF TEACHER

DAY 2: LEARNING OBJECTIVES

- Define efficiency as the ratio of useful work output to total work input in both simple and compound machines.
- Describe how friction, energy loss, and mechanical advantage reduce the performance of a machine.
- Perform efficiency calculations by using the standard formula to solve problems involving mechanical systems.

RECAP:

- What is a machine?
- What are the 2 categories of machine?
- What are the 6 types of simple machine?
- What is a compound machine?
- Give examples of compound machine.

FORCES ACTING ON A MACHINE

WEIGHT

TENSION

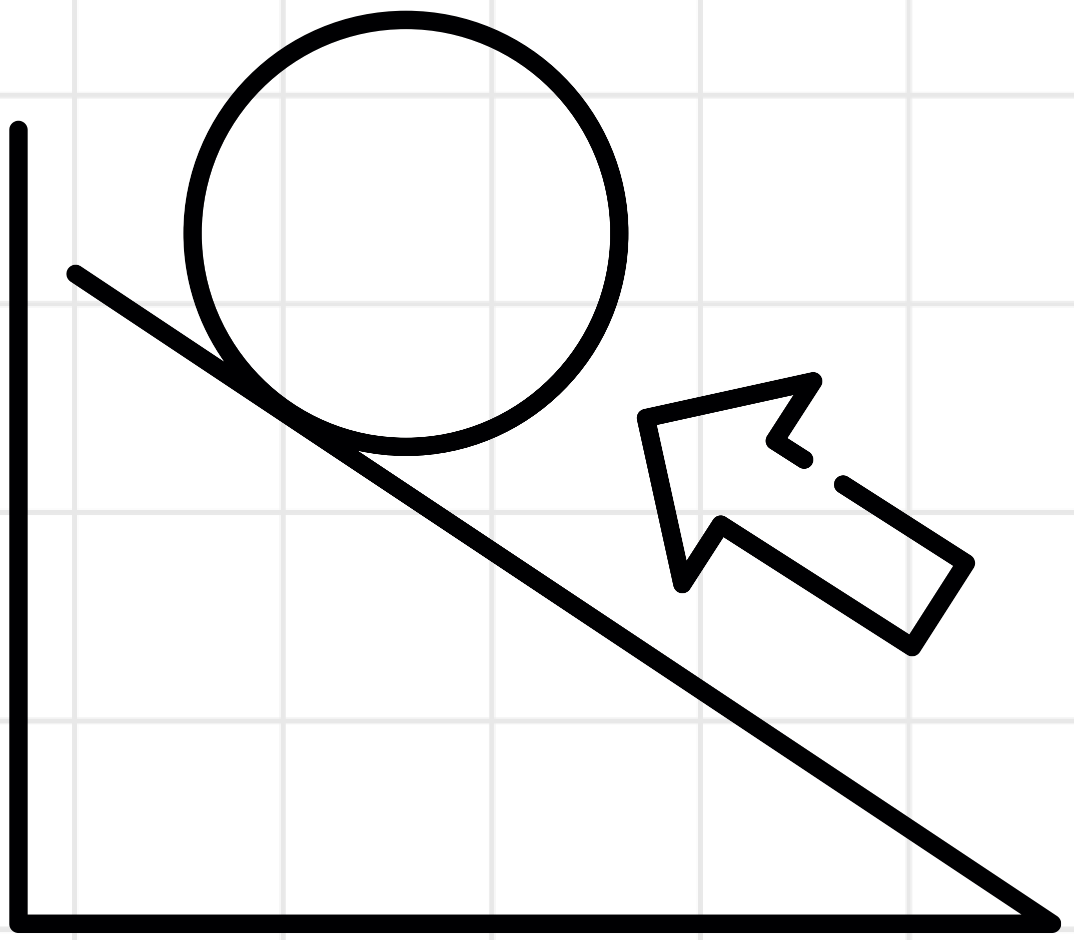
TORQUE

FRICTION

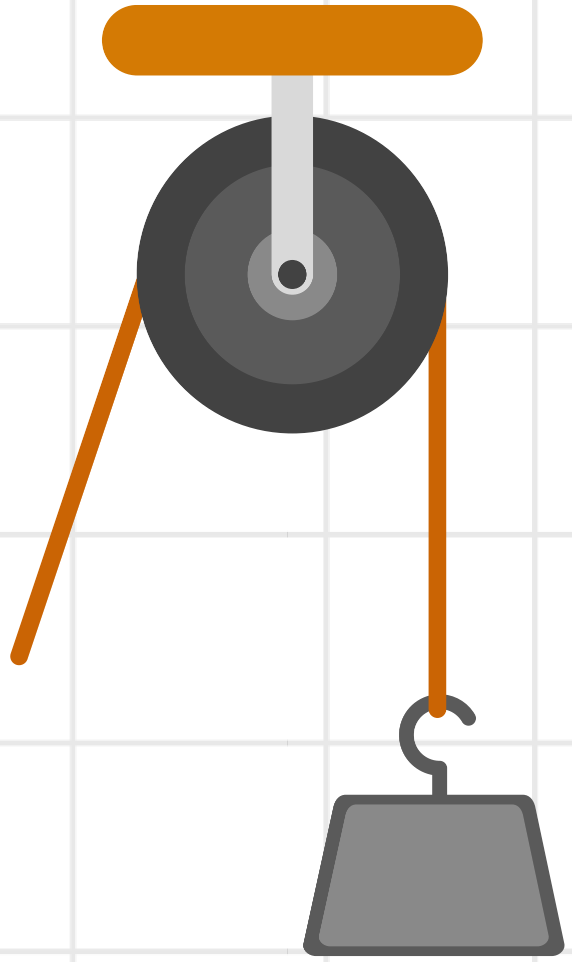
ACTIVITY: THE EFFICIENCY CHALLENGE

The Scenario

Imagine you are at a construction site and must move a heavy barrel from the ground to a platform above. You have three tools available:



RAMP (INCLINED PLANE)



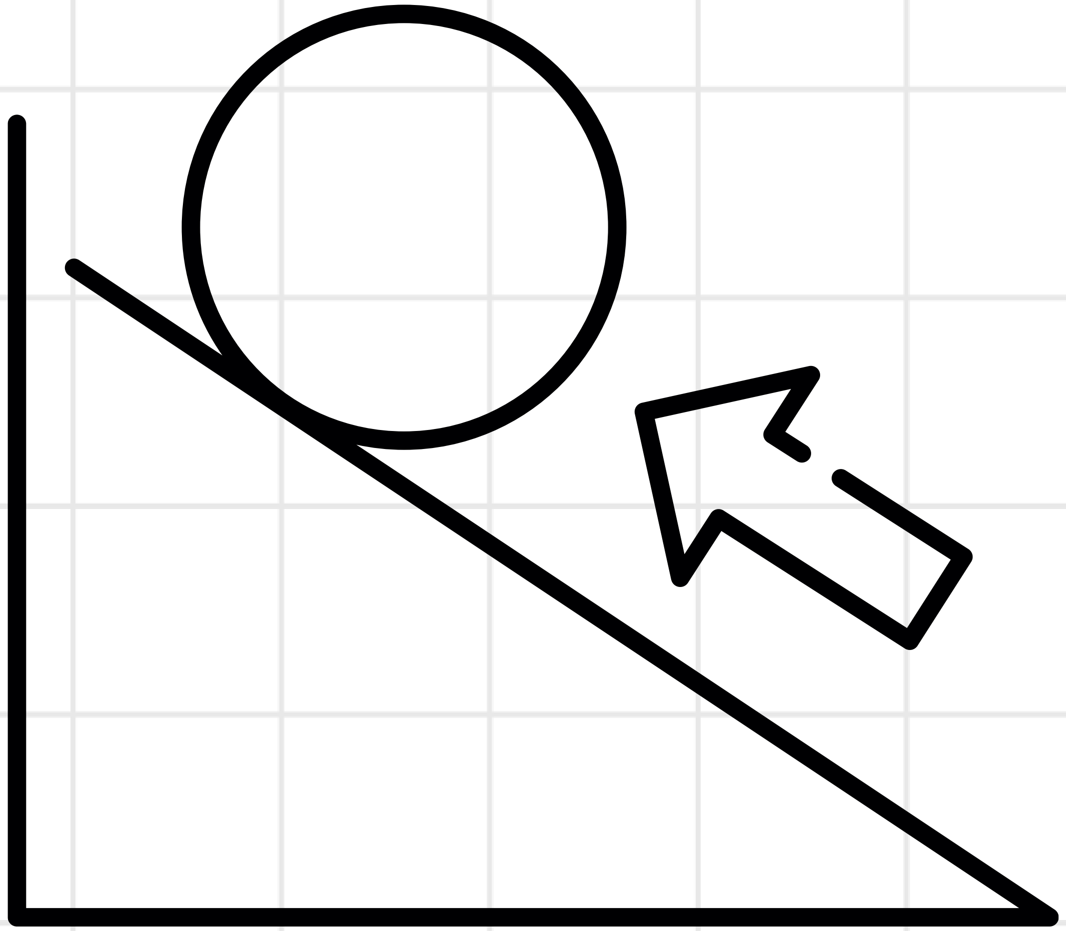
PULLEY SYSTEM

LEVER (CROWBAR)

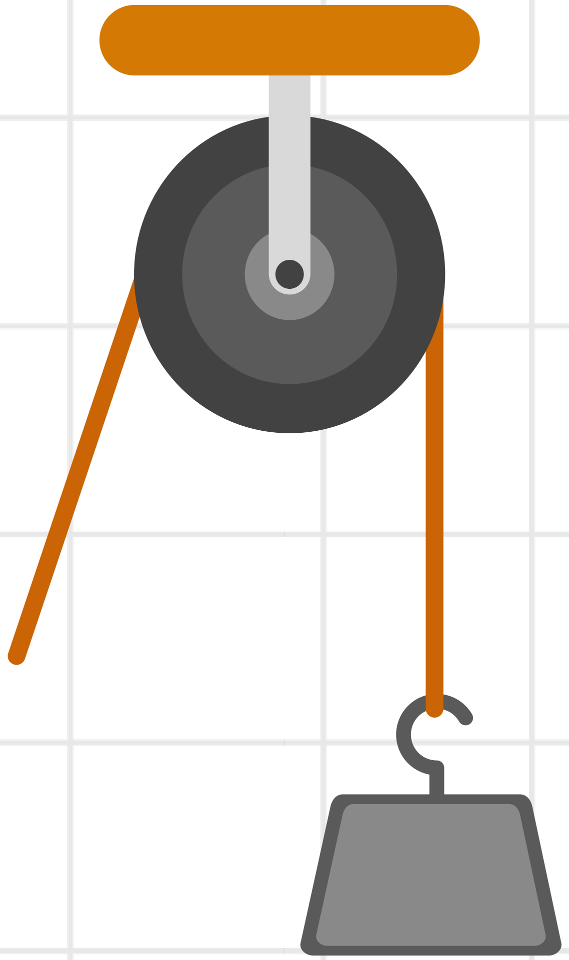
ACTIVITY: THE EFFICIENCY CHALLENGE

QUESTION:

1. Which of the three options would be the most efficient for this specific task?



RAMP (INCLINED PLANE)



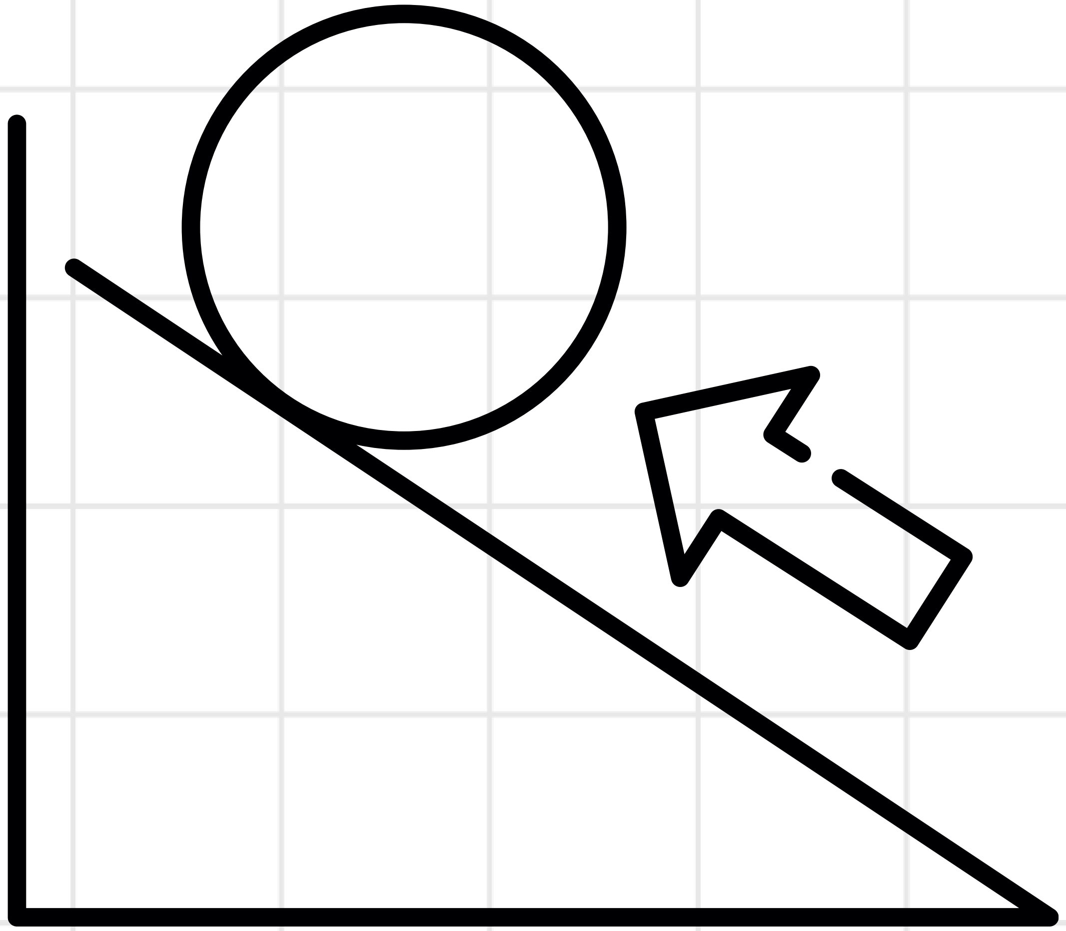
PULLEY SYSTEM

LEVER (CROWBAR)

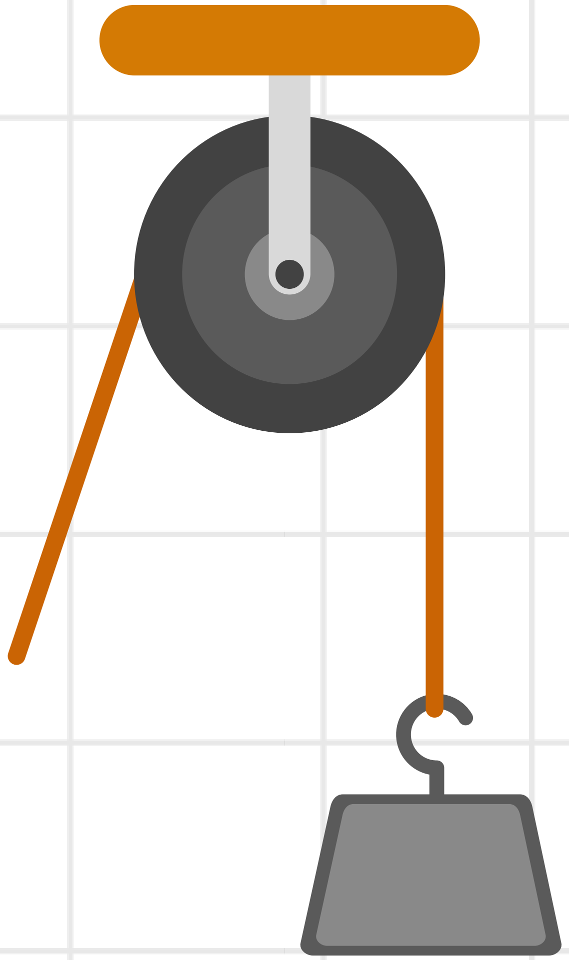
ACTIVITY: THE EFFICIENCY CHALLENGE

QUESTION:

2. How do forces like friction, gravity, and torque affect your choice?



RAMP (INCLINED PLANE)



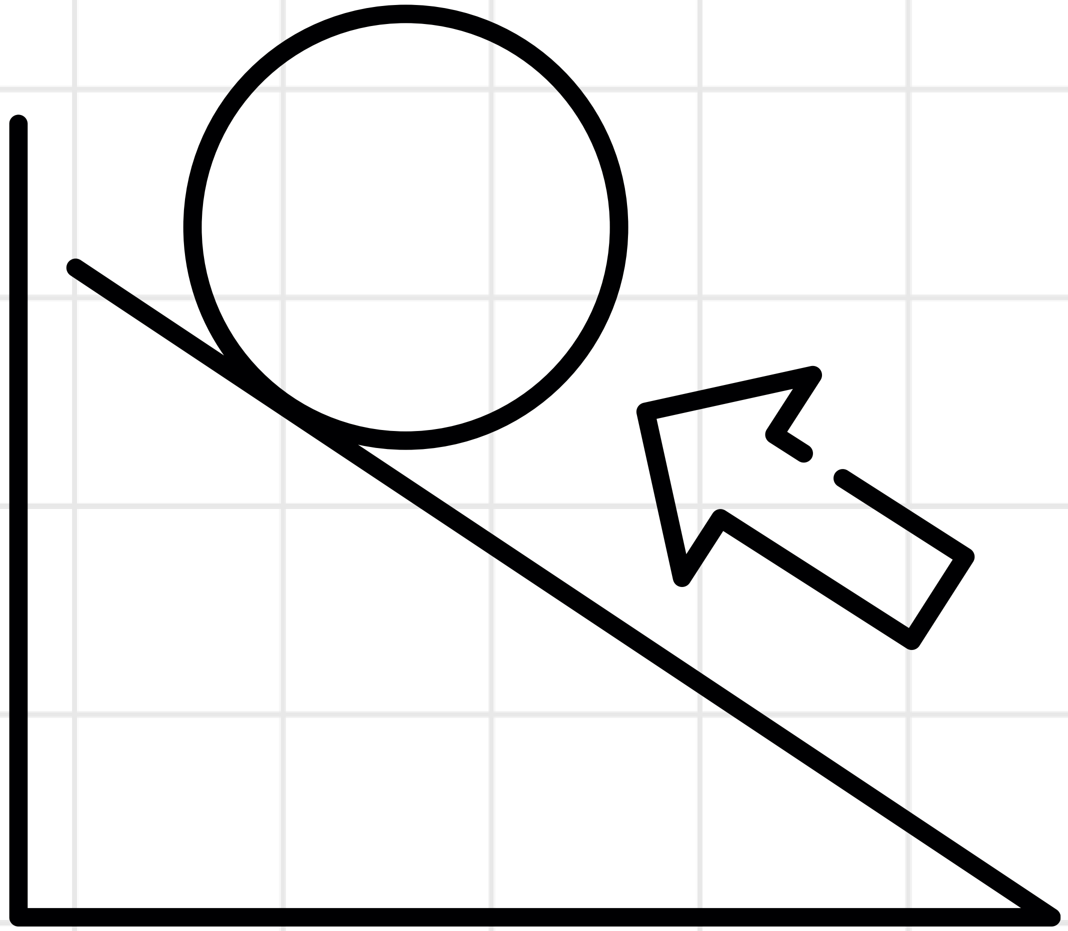
PULLEY SYSTEM

LEVER (CROWBAR)

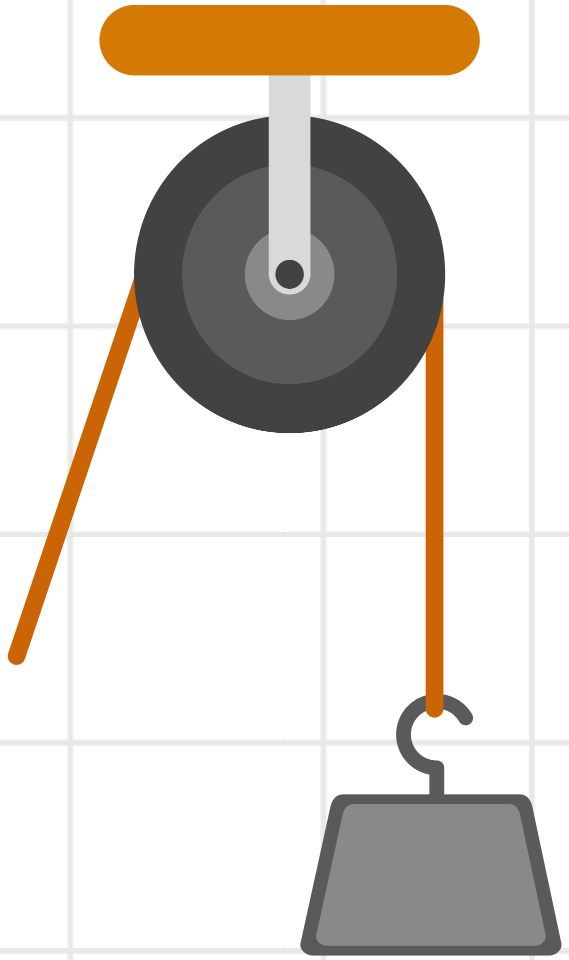
ACTIVITY: THE EFFICIENCY CHALLENGE

QUESTION:

3. Why is your chosen tool better for energy efficiency and ease of use compared to the others?



RAMP (INCLINED PLANE)



PULLEY SYSTEM

LEVER (CROWBAR)

MOTION & EFFICIENCY

- **Translational (Ramp):** You move the barrel in a straight line. Because the barrel rolls, you are converting your pushing energy directly into "height" (potential energy) with very little wasted heat.

- **Rotational (Pulley/Lever):** You have to rotate a wheel or a bar. Every time something rotates, there is a "rubbing" point (an axle or a fulcrum) where energy is lost to heat.

EFFICIENCY

Efficiency is a measure of how much of the energy you put into a machine actually goes toward doing the job, rather than being wasted.

Think of it like this:

- Input: The total energy or "effort" you give the machine.
- Output: The "useful work" the machine does for you.
- Waste: The energy lost to things like friction or heat.

A machine is considered "highly efficient" if it wastes very little energy and turns almost all of your effort into useful work.

EFFICIENCY IN SIMPLE MACHINES

Simple machines (like a lever or a ramp) usually have high efficiency.

- **Why:** They have very few moving parts.
- **The Result:** There are fewer surfaces rubbing together, which means there is less friction. Because friction is low, most of the energy you put in comes out as useful work.
- **Example:** A well-oiled pulley system might be 95% efficient.

EFFICIENCY IN COMPOUND MACHINES

Compound machines (like a bicycle or a car engine) usually have lower efficiency.

- **Why:** They are made by connecting many simple machines together.
- **The Result:** Each connection (where gears meet, or chains move) creates its own friction. Energy is lost at every single step.
- **Example:** A car engine is often only about 20% to 30% efficient because so much energy is lost to heat and friction among its hundreds of moving parts.

EFFICIENCY IN SIMPLE & COMPOUND MACHINES

Key Difference Summary

Machine Type	Number of Parts	Friction Level	Typical Efficie
Simple	One or very few	Low	**High**
Compound	Many connected parts	High	**Lower**

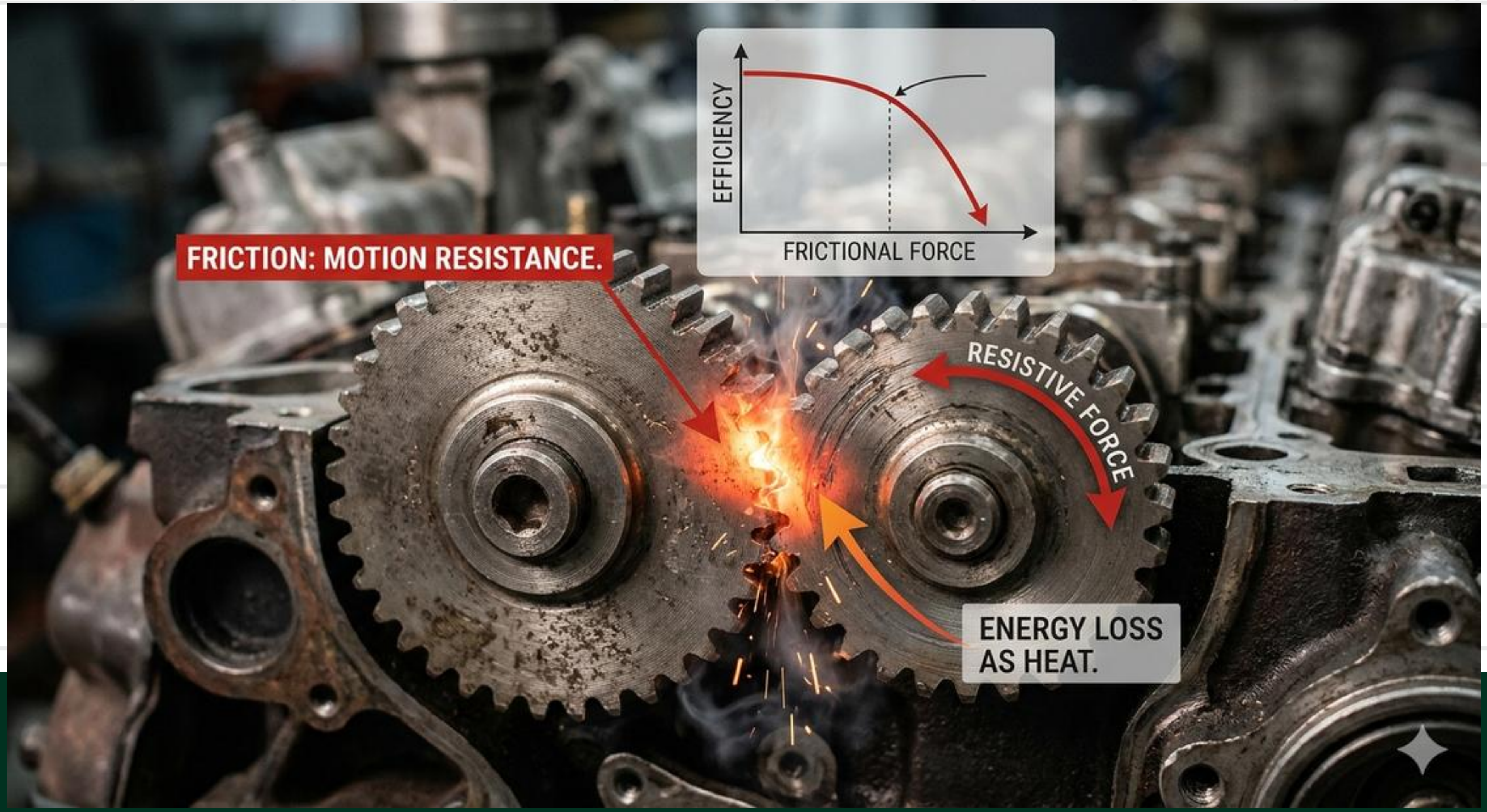
MECHANICAL ADVANTAGE

Mechanical Advantage is a measure of how much a machine multiplies your force to make a task easier. It acts as a strength multiplier, allowing you to move a heavy load with less effort by spreading the work over a longer distance.

FACTORS AFFECTING EFFICIENCY OF SIMPLE & COMPOUND MACHINE

1. FRICTION is the single biggest factor that reduces efficiency.

How it works: Whenever two surfaces rub against each other, they resist motion. This resistance requires extra work to overcome.

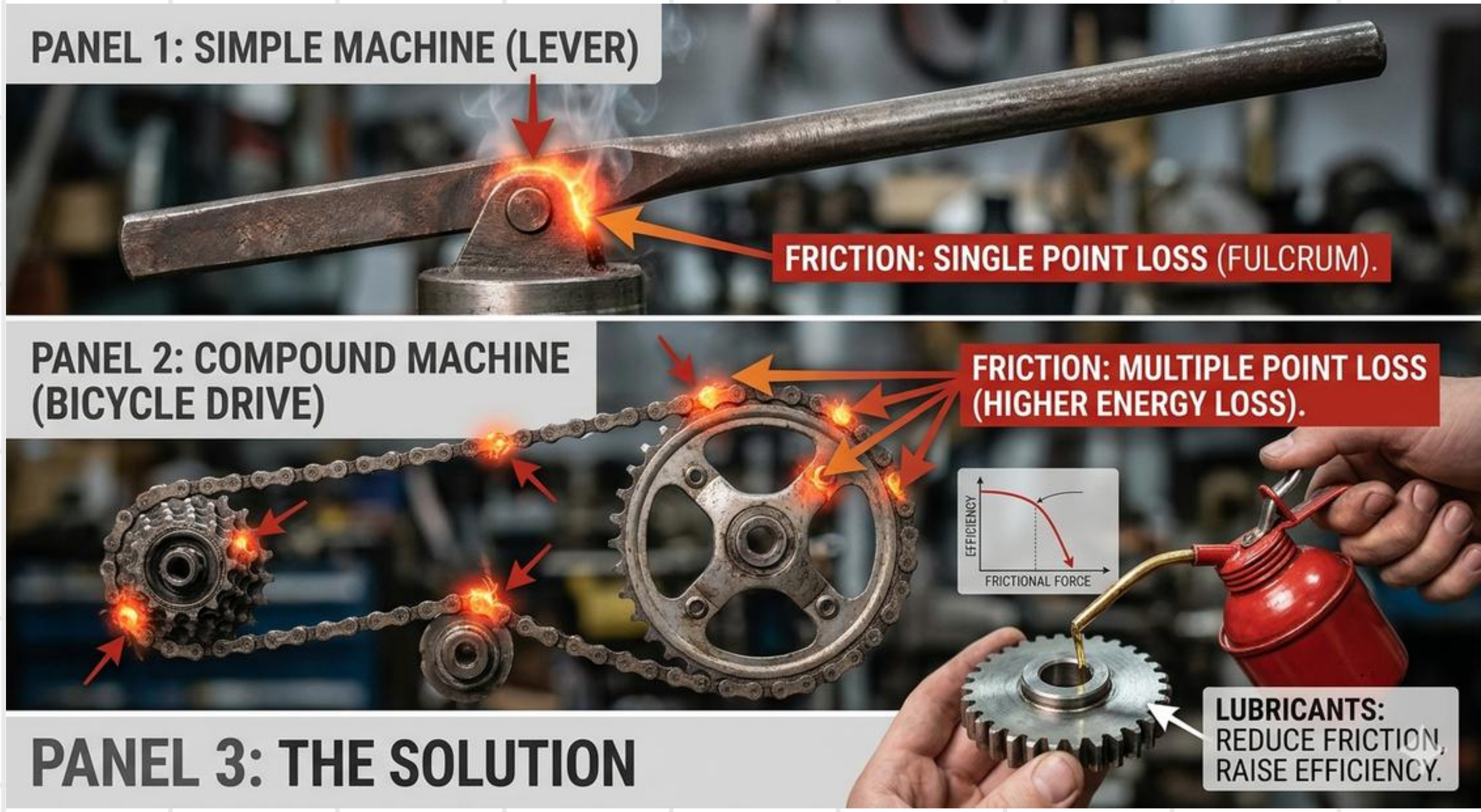


FACTORS AFFECTING EFFICIENCY OF SIMPLE & COMPOUND MACHINE

Simple vs. Compound:

- In simple machines, friction occurs at a single point (like a lever's fulcrum).
- In compound machines, friction happens at every connection point (gears, chains, and axles), leading to much higher energy loss.

Solution: Using lubricants like oil or grease reduces friction and raises efficiency.



FACTORS AFFECTING EFFICIENCY OF SIMPLE & COMPOUND MACHINE

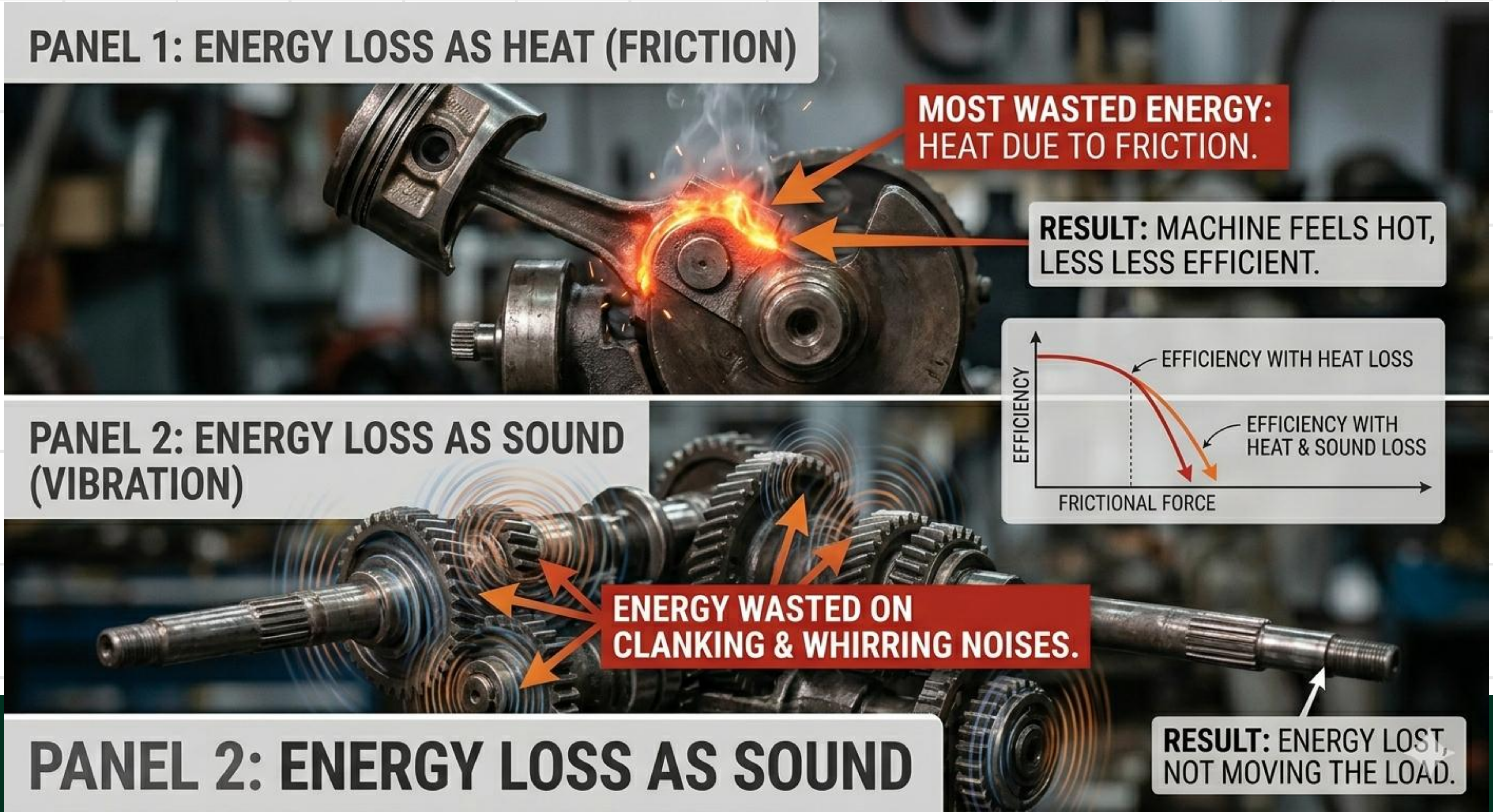
2. Energy Loss (Heat and Sound)

Energy cannot be destroyed, but it can be changed into forms that aren't useful for the task.

1. **Heat:** Most wasted energy in machines turns into heat due to friction. If a machine feels hot to the touch, it is being less efficient.
2. **Sound:** Energy used to create "clanking" or "whirring" noises is energy that isn't going into moving the load.

FACTORS AFFECTING EFFICIENCY OF SIMPLE & COMPOUND MACHINE

2. Energy Loss (Heat and Sound)



EFFICIENCY CALCULATION EXERCISE

The Scenario: Using a Ramp

Imagine you are pushing a crate that weighs 500 Newtons up a ramp. To get the crate to the top, you perform 2,000 Joules of work (Work Input). However, because the crate is only lifted to a certain height, the useful energy stored in the crate at the top is only 1,500 Joules (Work Output).

Formula:

$$\text{Efficiency} = \left(\frac{\text{Work Output}}{\text{Work Input}}\right) \times 100$$

EFFICIENCY CALCULATION EXERCISE

PRACTICE:

1. A pulley system requires you to do 400 Joules of work to lift a heavy bucket. After measuring the height and weight of the bucket, you find that the useful work done was 320 Joules.

EFFICIENCY CALCULATION EXERCISE

The Result:

The ramp is 75% efficient. This means 75% of your energy went into lifting the crate, while 25% was wasted (likely as heat caused by friction between the crate and the ramp).

EFFICIENCY CALCULATION EXERCISE

PRACTICE:

2. An electric motor uses 1,000 Joules of electrical energy to lift an elevator. However, the actual mechanical work done to move the elevator is only 850 Joules. The rest of the energy is lost as heat and sound.

Question: What is the efficiency of the motor?

EFFICIENCY CALCULATION EXERCISE

PRACTICE:

3. To pry open a heavy crate, you push down on a lever with 150 Joules of work. Because the lever is old and rusty, it creates a lot of friction at the fulcrum. The lever only delivers 105 Joules of work to the crate lid.

Question: Calculate the efficiency of the rusty lever.

Reflection: If you oiled the lever, would the Work Output increase or decrease?

QUIZ!

1. Which of the following best defines the efficiency of a machine?

A) The total amount of energy a machine can store.

B) The ratio of useful work output to the total work input.

C) The speed at which a machine performs a specific task.

D) The total weight a machine is capable of lifting.

QUIZ!

2. Which factor is most responsible for reducing the efficiency of a pulley system?

A) The length of the rope used.

B) The height to which the object is lifted.

C) Friction between the rope and the wheel.

D) The mass of the object being moved

QUIZ!

3. How does increasing the number of moving parts in a machine generally affect its efficiency?

A) It has no effect on efficiency.

B) It increases efficiency by distributing the load.

C) It makes the machine 100% efficient.

D)It decreases efficiency because more energy is lost to friction at each connection.

QUIZ!

4. If a lever requires 200 Joules of work input to produce 150 Joules of useful work output, what is its efficiency?

A) 50%   B) 75%   C) 80%   D) 133%

QUIZ!

5. A cyclist puts 500 Joules of energy into the pedals. Because of the friction in the chain and the gears, only 460 Joules of that energy actually reaches the rear wheel to move the bicycle forward.

Question: What is the efficiency of this bicycle's drivetrain?

THANK YOU!

I hope you learn something new today.